

Wage & Employment Dynamics

THE WED PROJECT



Annual Survey of Hours and Earnings linked to 2021 Census of England and Wales

**** Dataset In Development ****

Synthetic (Fake) Dataset – Version 0.1 (beta)

Abstract

The Wage and Employment Dynamics (WED) team are collaborating with the Office for National Statistics (ONS) to link the Annual Survey of Hours and Earnings (ASHE) to the 2021 Census of England and Wales (CEW21). The ASHE-CEW21 dataset is currently in development and is expected to become available in Summer 2026.

The ASHE-CEW21 Synthetic (Fake) Dataset aims to support users in understanding the future capabilities of the real dataset. It has been generated from public metadata and aims only to represent the structure of the real dataset; it does not seek to reproduce any of the moments from the real microdata. The synthetic dataset is expected to be most useful to those without any prior experience of using ASHE or the ASHE-CEW11 dataset, giving them a broad insight into the data structure of the real ASHE-CEW21 dataset before that becomes available.

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1. Introduction

The Wage and Employment Dynamics (WED) team are collaborating with the Office for National Statistics (ONS) to link the Annual Survey of Hours and Earnings (ASHE) to the 2021 Census of England and Wales (CEW21). The ASHE-CEW21 dataset is currently in development and is expected to become available in Summer 2026.

To support users in understanding the capabilities of the eventual ASHE-CEW21 dataset, the WED team are releasing a beta version of a synthetic (fake) dataset. This document explains the nature of this synthetic (fake) dataset and the process used to generate it.

The synthetic dataset is one part of a wider suite of resources for ASHE-CEW11:

- **ASHE-CEW21 User Guide (v0.1 beta):** Outlines the protocol being adopted by ONS to link the ASHE and CEW21 datasets and describes the expected features of the linked data. It also discusses connections between the ASHE-CEW21 linkage and an earlier project in which ASHE was linked to the 2011 Census of England and Wales (CEW11).
- **ASHE-CEW21 Data Dictionary (v0.1 beta):** The data dictionary lists the variables that are expected to be present on the final published dataset, with the exception of the geographic identifiers and variables relating to the linkage process itself (still the subject of discussion with the ONS Census and Data Engineering Teams).

2. Concept

Table 1 provides a basic insight into the expected data structure for ASHE-CEW21 (reproduced from the ASHE-CEW11 User Guide v0.1).

Table 1: Illustration of the structure of the ASHE-CEW21 dataset

ADR_ID_ASHE	SERNO_ASHE	YEAR_ASHE	Other ASHE variables	RESIDENT_ID_CEN	Other CEW21 variables
112233	123450	2020	xxx	778899	yyy
112233	123451	2020	xxx	778899	yyy
112233	123452	2021	xxx	778899	yyy
112244	123453	2020	xxx		
112244	123454	2021	xxx		
112244	123455	2022	xxx		
112255	123456	2019	xxx	667788	yyy
112255	123457	2020	xxx	667788	yyy
112255	123458	2021	xxx	667788	yyy

Note: each ASHE respondent shown here may also be observed in other years of ASHE in the range 1999 to 2022.

Version 0.1 of the synthetic (fake) dataset provides a more extensive insight into the structure of the final dataset. It has been generated from public metadata (Version 0.1 of

the ASHE-CEW21 Data Dictionary) and so contains most of the ASHE and CEW21 variables that we expect to appear on the final dataset, with the exception of geographic identifiers and linkage variables (see above).

The synthetic dataset aims for **structural fidelity** with the eventual ASHE-CEW21 dataset – subject to the important caveats outlined in Section 3. It does not aim for any level of statistical fidelity. The data values contained within the synthetic dataset are fake and have been generated at random. This means that the dataset does not seek to reproduce any of the moments from the real microdata and **cannot** be used for any analytical purpose. Furthermore, it does not provide any guidance on the number of linked records or the size of sub-populations within the real dataset, and it cannot be used to identify relationships between variables.

The synthetic dataset is expected to be most useful to those without any prior experience of using ASHE or the ASHE-CEW11 dataset.¹ It will give novice users a broad insight into the expected data structure for ASHE-CEW21 before it becomes available.

Once the ASHE-CEW21 dataset has completed its development in Summer 2026, we will seek to release a second synthetic dataset which has a greater level of structural fidelity with the real data, and which aims for statistical fidelity across a small number of variables. This synthetic (demonstrator) dataset will be used to illustrate the analytical capabilities of the real dataset during training activities. Such demonstrations cannot utilise the real dataset, as disclosure controls prevent its removal from the Secure Research Service (SRS) environment.

3. Key properties of the synthetic dataset

The synthetic dataset has been generated from Version 0.1 of the ASHE-CEW21 Data Dictionary. It contains **only** those variables belonging to the main dataset; it **does not** contain any variables from the Household table or Relationship table. Moreover, some variables in the main dataset have been omitted for reasons of expediency. The variables that are included in the synthetic dataset are identified with value 1 in column G of the Data Dictionary worksheet (the column titled ‘synthetic_fake’).

The synthetic dataset contains far **fewer** records than the real dataset, again for reasons of expediency.

3.1. Synthetic ASHE records

The synthetic dataset contains 1,000 ASHE job records for every year from 1999-2022 (the date range that will be covered by the real dataset) – a total of around 24,000 records.

¹ Forth J, Phan V, Stokes L, Ritchie F, Whittard D, Bryson A and Singleton C (2022a) [ASHE-Census 2011 Data Linkage: User Guide for Drop 2 of the ASHE-Census 2011 Dataset](#). Bristol: Wage and Employment Dynamics Project.

This is considerably fewer than the number of records in the real ASHE dataset, which typically has around 150,000 job records per year. This is for reasons of expediency, given that there is no intention to reproduce any of the moments from the real dataset. Years are identified on the variable YEAR_ASHE.

In each year, job records are uniquely identified by the variable SERNO_ASHE. The variable contains generated values running from 1 to 1,000 (the total number of observations within the year). Individuals are uniquely identified by the variable ADR_ID_ASHE, which contains generated values running from 1 to 950. As in the real dataset, a minority of individuals have multiple job records in a given year. As implied by the values stated above, the number of individuals with multiple job records in a given year has been arbitrarily set at 50. As in the real dataset, individuals with multiple job records in a given year can be identified via MJOB_ASHE and DJOB_ASHE.

The synthetic dataset is formed as a balanced panel of ASHE individuals across years. In other words, each ADR_ID_ASHE value (from 1 to 950) is present in every year of the data from 1999-2000. This is **not** the case in the real ASHE dataset, where ASHE sample members may be present in varying durations, and those users wishing to conduct longitudinal analysis at the level of the individual must deal with panel attrition.²

All other ASHE numeric variables in the synthetic dataset have been populated with a random number generator. All generated values are integers. Where the underlying variable is categorical (e.g. ADR_ASHE, APP_ASHE, ASHEFURLOUGH_ASHE) the generated values have been constrained to fit the list of categories specified in the Data Dictionary. For example, the variable ADR_ASHE holds the values 1 or 2 at random. Where the variable is continuous (e.g. ACTHRS_ASHE, AGE_ASHE, AGP_ASHE), generated values have been constrained to lie within a reasonable (but ultimately arbitrary) range. For example, the variable ACTHRS_ASHE has values between 1 and 40, chosen at random.

All ASHE string variables in the synthetic dataset contain a string indicating the *format* of the real data. For instance, the variable APPDATE_ASHE contains the string “MMYYYY”. There has been no attempt to populate string variables with artificial values.

Logical patterns of missing data that exist in the real ASHE dataset are generally replicated in the synthetic dataset. For instance, the variables introduced into ASHE in 2021 to capture furlough are only observed in that year; the SIC03 variable is only observed in the years 1999-2008 and the SIC07 variable is only observed in the years 2008-2022. However, users are urged to consult the ONS ASHE User Guide for authoritative guidance on variable availability across the years of the real ASHE dataset.

Otherwise, there has been no attempt to enforce logical consistency between the ASHE variables in the synthetic dataset. For instance, the variable recording a job’s hourly pay

² Forth J, Phan V, Stokes L, Ritchie F, Whittard D, Bryson A and Singleton C (2022b) [Longitudinal Attrition in ASHE](#). Bristol: Wage and Employment Dynamics Project.

will not hold a value that corresponds to the ratio of weekly pay to weekly hours (unless by chance).

3.2. Synthetic CEW21 records

Synthetic CEW21 records have been generated at the person level for a single year (2021). Individual persons are uniquely identified by the variable RESIDENT_CEN, which contains generated values running from 1 to 950.

CEW21 variables have been populated using the same principles as outlined above in respect of ASHE. Specifically:

- all CEW21 numeric variables in the synthetic dataset have been populated with integers using a random number generator
- the values in categorical variables have been constrained to fit the list of categories specified in the Data Dictionary
- the values in continuous variables have been constrained to lie within a reasonable (but ultimately arbitrary) range
- string variables contain a string indicating the *format* of the real data, rather than artificial values
- logical patterns of missing data (e.g. arising from questionnaire routing in CEW21) have been replicated
- there has been no attempt to enforce logical consistency between the CEW21 variables or between the ASHE and CEW21 variables (any such consistency arises purely by chance).

As noted earlier, the synthetic dataset only contains CEW21 variables at the individual-level. It does not contain any variables from the Household table or Relationship table.

3.3. ASHE-Census linkage in the synthetic dataset

The synthetic ASHE data has been linked to the synthetic CEW21 data at the person level by matching ADR_ID_ASHE to RESIDENT_ID_CEN. Since ADR_ID_ASHE and RESIDENT_ID_CEN are both indexed from 1 to 950, this would ordinarily result in CEW21 data being linked to each and every ASHE record in the synthetic dataset. However, to illustrate the fact that the eventual linkage rate between the real ASHE and the real CEW21 will be partial, 20% of the linkages have been manually blocked. These 20% of ASHE records will have no CEW21 data linked to them. The linkage rate of 20% has been set arbitrarily. The linkage rate for ASHE-CEW11 was 74%³ but the linkage rate for ASHE-CEW21 is not yet known.

³ Forth J, Phan V, Stokes L, Ritchie F, Whittard D, Bryson A and Singleton C (2022a) [ASHE-Census 2011 Data Linkage: User Guide for Drop 2 of the ASHE-Census 2011 Dataset](#). Bristol: Wage and Employment Dynamics Project.

In the real dataset, there will be no numeric relationship between ADR_ID_ASHE and RESIDENT_ID_CEN. ASHE records will be linked to CEW21 via the ONS Demographic Index.

4. Data formats

The ASHE-CEW21 synthetic (fake) dataset is available in Stata (.dta) format and Comma Separated Value (.csv) format.

Variables are labelled in the Stata dataset. Users of the CSV dataset should refer to v0.1 of the ASHE-CEW21 Data Dictionary for variable labels.

Neither dataset incorporates value labels. Code lists for categorical variables are provided in the Data Dictionary.

5. Cautionary notes

The synthetic dataset aims for **structural fidelity** with the eventual ASHE-CEW21 dataset – subject to the important caveats outlined in Section 3.

It does not aim for any level of statistical fidelity. The data values contained within the synthetic dataset are fake. As a consequence, the synthetic dataset **cannot** be used for any analytical purpose. It does not provide any guidance on the size of sub-populations within the real dataset and it cannot be used to identify relationships between variables.

6. Further information

All documentation relating to the ASHE-CEW21 linkage may be obtained from the website of the WED project: <https://www.wagedynamics.com/>

For clarifications and to provide feedback on the synthetic dataset please contact: wagedynamics@uwe.ac.uk